

Ob-
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Ser-
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In-
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De-
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Ser-
vice
ID
User
Story
ID
User
Story
Item
User
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Tech-
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Story
Est.
Item
User
Story
Tech-
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cal
Story
Est.
YOLOInferenceEngine
DetectedObject
FFmpegFrameCapture
SQLAlchemyDetectionRepository
DetectionEvent
camera_id
created_at
DetectionEvent

RegionOfInterest
{x,
y}
SQLAlchemyROIRepository
severity
schedule
RegionOfInterest
ROIEvaluator
IntrusionDetectionUseCase
IntrusionEvent
SQLAlchemyIntrusionRepository
IntrusionEvent
EmailAlertService
PATCH
/alerts/{id}/acknowledge
status
ACKNOWLEDGED
acknowledged_by
acknowledged_at
resolution_note
Alert
GET
/intrusion-events
camera_id
severity
roi_id
??
Cam-
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Owner
Cam-
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Owner
AI
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Run
Ob-
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FFmpegFrameCaptureYOLOInferenceEngineDetectedObjectRunDetectionUseCaseDetection_EventIDDetectionReposi
Man-
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IntrusionEvent
Ap-
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Layer
RunDetectionUseCase
IntrusionDetectionUseCase
AlertOrchestrator

In-
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truc-
ture
Layer
YOLOInferenceEngine
FFmpegFrameCapture
ROIEvaluator

API
Layer
Sequence

1
Ob-
ject
De-
tec-
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Work-
flow
??

FFmpegFrameCapture
RunDetectionUseCase
YOLOInferenceEngine
DetectedObject
DetectionEvent
Sequence

2
In-
tru-
sion
De-
tec-
tion
Work-
flow
??

DetectionEvent
RegionOfInterest
ROIEvaluator
IntrusionEvent
AlertOrchestrator
Sequence

3
In-
tru-
sion
Alert
Work-
flow
??

IntrusionEvent
AlertOrchestrator
EmailAlertService
Alert
PENDING
PATCH
ACKNOWLEDGED
yolo11n
x
yolo11m
yolo11l
IYOLOInferenceEngine
asyncio
asyncio.Queue
tracker=bytetrack.yaml
ROIEvaluator

per-
son-
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tered
zone
X
track

42
has
been
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ously
present

in
zone
X
for

3
sec-
onds
aioredis
intrusion-events
send json
IYOLOInferenceEngine
camera_id
created_at
intrusion_events
severity
IN

('HIGH',